

Gaston Elia Bujia

Data Scientist | Machine Learning | Predictive Modeling | Python | Computational Statistics

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Data Scientist with a strong background in mathematics, machine learning, and applied research. I combine hands-on experience building and shipping predictive models, including a production-deployed clinical scoring system, with doctoral training in computational modeling, Bayesian methods, and deep learning. I am especially interested in roles where rigorous quantitative thinking, Python-based analysis, and clear communication can turn complex data into actionable decisions.

Relevant Experience

Research Data Scientist Everything ALS | *March 2024 - Present*

- Developed a production-deployed dysarthria severity scoring model using Whisper, PyTorch, and scikit-learn, integrated into clinical trial workflows. Built predictive models using accelerometer and voice signals for ALS disease progression monitoring.
- Applied survival analysis and collaborated with clinical scientists to translate research questions into ML solutions for complex clinical datasets.

Graduate Researcher Applied Artificial Intelligence Lab (UBA-CONICET) | *2018 - 2026*

- Designed and evaluated end-to-end ML pipelines for predicting sequential human behavior from real experimental datasets, combining deep learning, Bayesian modeling, CNN-based saliency models, and reproducible analysis workflows.
 - Contributed to **ViSioNS** (NeurIPS 2022), an open-source benchmark for evaluating computational models, analogous to model evaluation frameworks used in production ML.
 - Developed **sIBS**, a probabilistic visual-search model combining Bayesian inference and saliency maps, published in *Frontiers in Systems Neuroscience*.
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Key Projects

- **Dysarthria scoring model (Everything ALS):** Production-deployed clinical NLP/audio model built with Whisper, PyTorch, and scikit-learn.
 - **ViSioNS (NeurIPS 2022):** Open-source benchmark for computational models of human visual search, adopted by the research community.
 - **Recommendation systems (B.Sc. thesis):** Implemented Factorization Machines to address the cold-start problem in low-data recommendation scenarios.
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Core Skills

- **ML and data stack:** Python, NumPy, Pandas, scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, Weights & Biases.
 - **Modeling:** Predictive modeling, deep learning, NLP/audio models, survival analysis, Bayesian modeling, linear mixed-effects models.
 - **Workflow:** Git/GitHub, Docker basics, Jupyter, Matplotlib, Seaborn, Plotly, technical writing, cross-functional collaboration.
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Teaching & Publications

- **Teaching:** Machine Learning and Computational Statistics at UBA; graduate ML teaching in the Data Mining and Knowledge Discovery master's program of the Department of Computer Science, FCEN-UBA.
- **Mentoring:** undergraduate thesis co-supervision and research mentorship in computational modeling and behavioral data.
- **Publications:** Several publications in peer-reviewed scientific journals and conferences, including Scientific Reports, NeurIPS, and Frontiers in Systems Neuroscience.